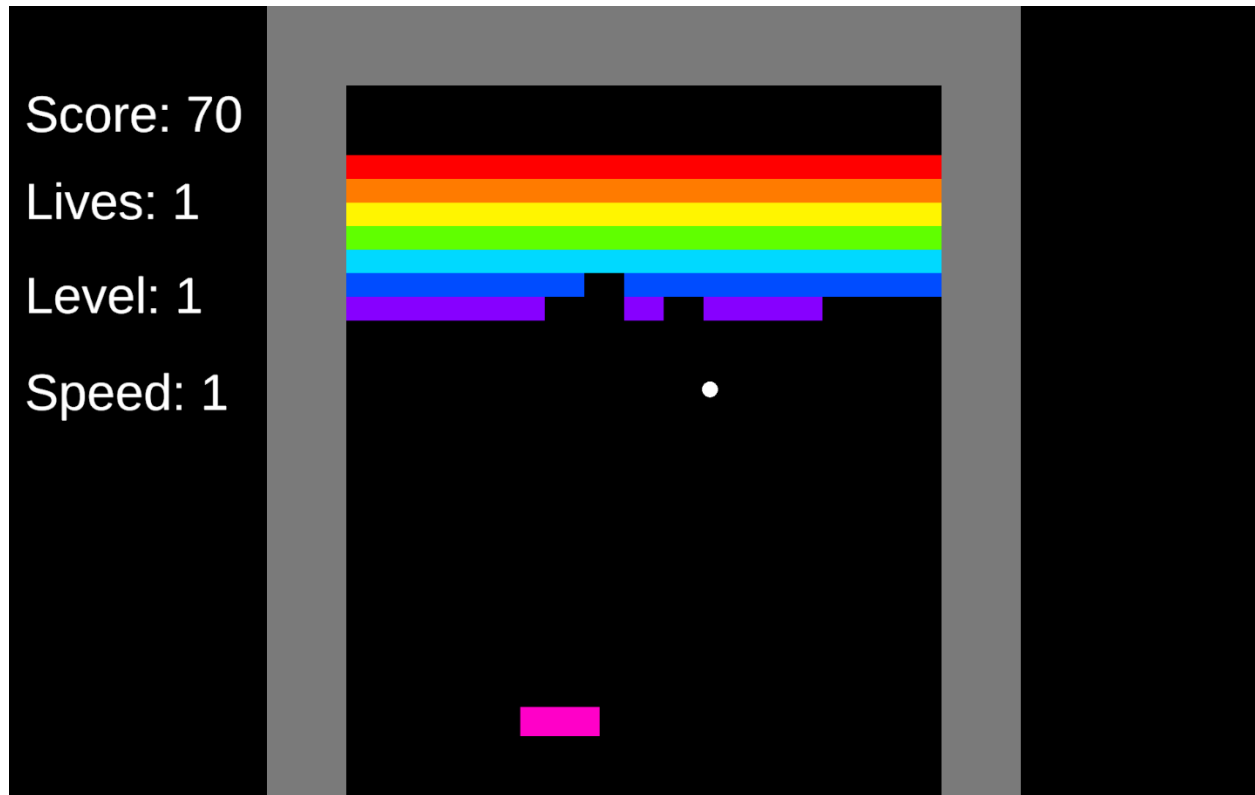


Clean-Up By Ryan Mayleas

Concept



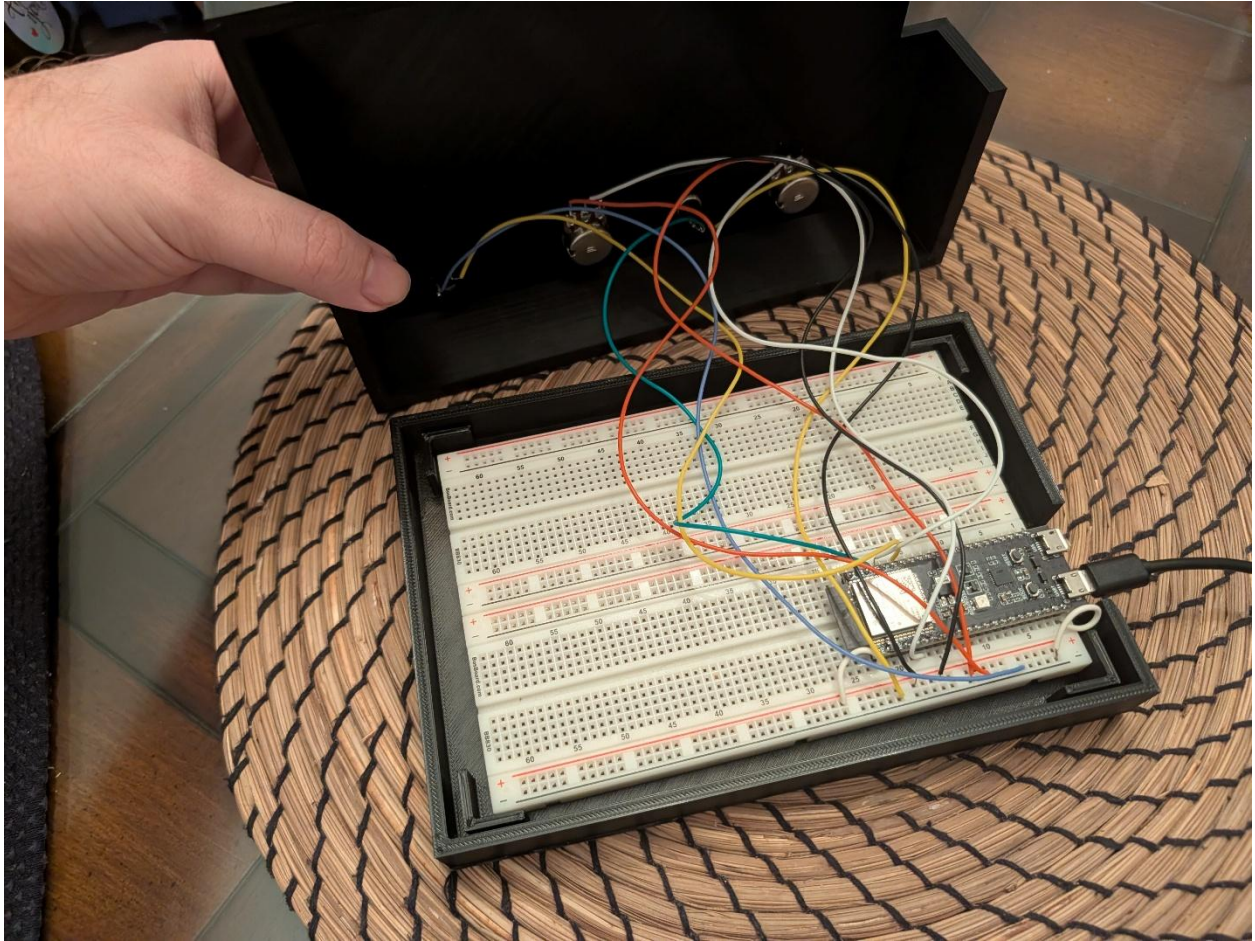
My project is a recreation of the 1976 video game Breakout that is controlled by a microcontroller connected to some hardware sensors. In the game, you control a paddle that has to clear out a pile of bricks by knocking a ball towards them so they break. However, they must make sure the ball doesn't go past the paddle or else they will lose it, lose all balls and the game is over. As the level goes on, the ball will end up getting faster and breaking all the bricks brings you to the next level. A major difference between this recreation and the original game is that the player can increase the speed in which the ball moves not only to increase the difficulty, but also to increase the number of points they can get.

How to Play



This project uses two potentiometers and one button to control the game. The left potentiometer controls the paddle's position, the button brings a ball into play if there isn't one or restarts the game if it's over, and the right potentiometer adjusts the speed of the ball (Note: Ball speed cannot be adjusted if the ball is in play). A piezo buzzer is also used to provide sound effects for when the ball knocks into the paddle, a wall, or a brick, or if the ball is goes out of bounds.

How it Works



All of the hardware sensors and the buzzer are connected to an ESP32 microcontroller through a breadboard and wires that were soldered onto the components. The components (except for the buzzer) are bolted onto the top of a 3d printed case that also houses the microcontroller and the breadboard. The software was created using Unity, a free video game engine, and Ardity, a library for Unity that allows it to connect to the microcontroller over a serial port. The microcontroller sends data that it reads from the sensors to the software, and the software applies that data to the objects within the game. The software also sometimes sends data back to the microcontroller, and the microcontroller uses that data to tell the buzzer what tone to play and for how long.